Laboratory Activity 6

Use Case Diagram Development - CategorAIze Email Intelligence System

Part 1: Project Scope and Actor Identification

This diagram follows the approved project scope in the team's activity worksheet: Gmail connection, email categorization and prioritization, analysis and deadline review, notifications, search and cleanup, reply drafting, and Google Calendar availability and deadline export. The web app, background processing, data store, and NLP capabilities are inside the system boundary.

Students, professionals, and other people who manage email use the same mailbox features and permissions in this project scope. They are represented by one Email User actor. Organization administration, deployment administration, and other inherited application features are outside this approved project scope.

Actor Inventory

Actor	Description and Main Interaction
Email User (Primary Actor)	A person, such as a student or professional, who connects and manages their own Gmail mailbox. Connects the account, creates custom categories, reviews categorized and prioritized email, checks summaries and deadlines, views notifications, searches and cleans email, drafts replies, and corrects classifications.
Gmail Service	External email and OAuth service. Authorizes mailbox access, supplies messages and updates, stores drafts, sends approved replies, and applies requested mailbox changes.
Google Calendar Service	External calendar service connected with the user's consent. Supplies availability and creates calendar events after the user confirms an export.

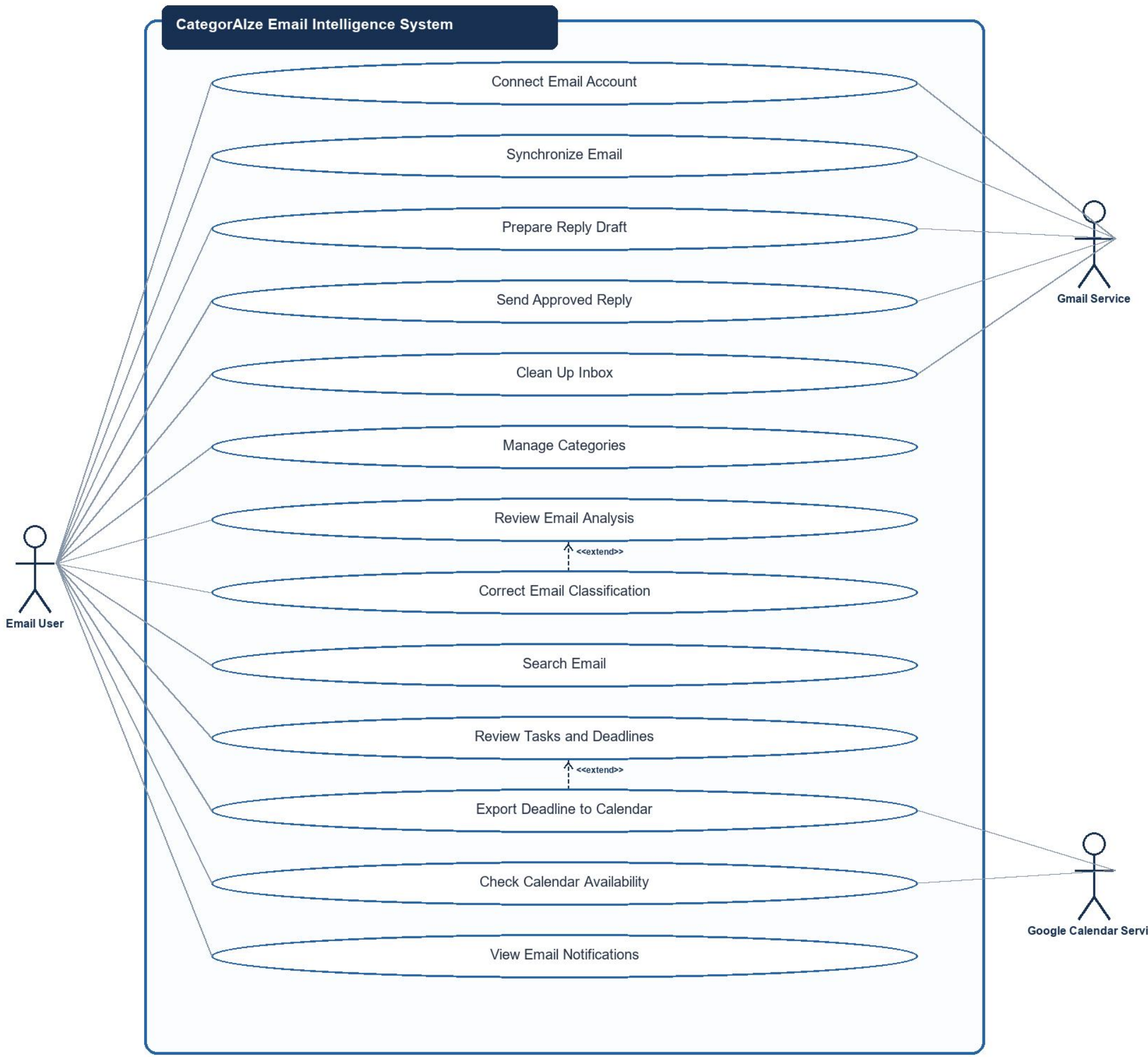
Part 2: Use Case Identification

Actor, use case, and purpose inventory

Actor	Use Case	Purpose
Email User	Connect Email Account	Allows the user to sign in and authorize CategorAIze to access Gmail.
Gmail Service	Connect Email Account	Provides OAuth authorization and mailbox access.
Email User	Synchronize Email	Allows the user to retrieve new and updated messages for processing and review.
Gmail Service	Synchronize Email	Supplies new and updated messages from the connected mailbox.
Email User	Manage Categories	Allows the user to create, edit, activate, or remove categories and set classification preferences.
Email User	Review Email Analysis	Allows the user to inspect category predictions, priority explanations, summaries, suggested actions, and deadlines when available.
Email User	Correct Email Classification	Allows the user to correct an email's category to improve future predictions.
Email User	Search Email	Allows the user to find relevant messages using semantic search and available filters.
Email User	Review Tasks and Deadlines	Allows the user to review extracted actions, approaching deadlines, and important dates.
Email User	Prepare Reply Draft	Allows the user to generate and edit a contextual reply before saving it as a Gmail draft.
Gmail Service	Prepare Reply Draft	Stores the reply in Gmail Drafts when requested.
Email User	Check Calendar Availability	Allows the user to check available dates and times for scheduling.
Google Calendar Service	Check Calendar Availability	Provides availability information from the connected calendar.
Email User	Send Approved Reply	Allows the user to send a reviewed reply after confirming it.
Gmail Service	Send Approved Reply	Sends the reply after receiving the user's confirmed request.
Email User	Clean Up Inbox	Allows the user to archive selected messages, unsubscribe from unwanted emails, or configure unwanted-email blocking.
Gmail Service	Clean Up Inbox	Applies requested mailbox changes and supports unsubscribe actions.
Email User	Export Deadline to Calendar	Allows the user to review and confirm an extracted deadline before creating a calendar event.
Google Calendar Service	Export Deadline to Calendar	Creates a calendar event after the user confirms its details.
Email User	View Email Notifications	Allows the user to view notifications about important messages and upcoming deadlines.

Part 3: UML Use Case Diagram

CategorAIze Email Intelligence System



Dashed <<extend>> arrows point from an optional use case to its base use case.

Parts 4 and 5: Peer Review and Revision

CategorAIze Email Intelligence System

Peer Review Checklist

[x] System boundary is clearly labeled.

[x] Actors are outside the system boundary.

[x] Use cases are inside the system boundary.

[x] Use case names describe meaningful system services.

[x] Actors are connected to the appropriate use cases.

[x] Diagram reflects the approved project scope.

[x] Major user interactions are represented.

[x] No obvious missing or contradictory interactions remain after revision.

[x] Diagram is readable and properly organized after spacing revisions.

Two Strengths

1. The diagram is clear and well organized, with the system boundary, actors, and use cases properly labeled.
2. It shows important functions, including email categorization, email analysis, and task and deadline management, as well as the Gmail and Google Calendar integrations.

Recommendations and Missing Use Case

1. Improve spacing between use cases and association lines to make the diagram easier to read.
2. Show that Correct Email Classification is an optional action during Review Email Analysis, and that Export Deadline to Calendar is an optional action during Review Tasks and Deadlines, using <<extend>> relationships.

Potentially missing: UC13, View Email Notifications, lets the user view notices about important messages and upcoming deadlines. Added to the inventory and revised diagram.

Part 5: Revision Notes

1. Improved spacing between use cases and association lines.
2. Added <<extend>> from Correct Email Classification to Review Email Analysis, and from Export Deadline to Calendar to Review Tasks and Deadlines. Each extension is an optional action in the related review flow.
3. Added UC13, View Email Notifications, for notices about important email and upcoming deadlines.